

Week-2 Module

Program 1: Linear Search

```
import java.util.Scanner;
```

```
public class LinearSearch {  
    public static void main (String [] args) {  
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();  
        int[] arr = new int [n];  
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }
```

```
        int x = sc.nextInt();  
        int index = -1;  
        for (int i = 0; i < n; i++) {  
            if (arr[i] == x) {  
                index = i;  
                break;  
            }  
        }
```

```
        if (index != -1)
```

```
            System.out.println("Found at index" + index);
```

```
        else
```

```
            System.out.println("Not Found");  
        }  
    }  
}
```

DATE: / /
PAGE: /
Sample input

5

101 105 113 110 130

110

Output

Found at index 3

Program 2: Binary search (Product ID)

```
import java.util.Scanner;
```

```
public class Binary Search {
```

```
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int [n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        int target = sc.nextInt();
```

```
        int low = 0;
```

```
        int high = n - 1;
```

```
        boolean found = false;
```

```
        while (low <= high) {
```

```
            int mid = (low + high) / 2;
```

```
            if (arr[mid] == target) {
```

```
                found = true;
```

```
        break;
    }
    else if (arr[mid] < target)
        low = mid + 1;
    else
        high = mid - 1;
    }
    if (found)
        System.out.println("Product available");
    else
        System.out.println("Product Not available");
```

Output

Product Available

DATE: / /
PAGE: /
Program 3: Bubble Sort (Exam Score)

Java Code

```
import java.util.Scanner;

public class BubbleSort {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        for (int i = 0; i < n-1; i++) {
            for (int j = 0; j < n-i-1; j++) {
                if (arr[j] > arr[j+1]) {
                    int temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp;
                }
            }
        }

        for (int num : arr) {
            System.out.print(num + " ");
        }
    }
}
```

Program 4: Insertion Sort (Library Book No.)

```
import java.util.Scanner;
```

```
public class Insertion Sort {  
    public static void main (String[] args) {  
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();  
        int [] arr = new int [n];
```

```
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }
```

```
        for (int i = 1; i < n; i++) {  
            int key = arr[i];  
            int j = i - 1;
```

```
            while (j >= 0 && arr[j] > key) {  
                arr[j+1] = arr[j];  
                j--;
```

```
            }  
            arr[j+1] = key;
```

```
        }  
        for (int num : arr) {  
            System.out.print (num + " ");
```

```
        }  
    }  
}
```

Program 5: Selection Sort (Player Scores)

```
import java.util.Scanner;

public class SelectionSort {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);

        int n = sc.nextInt();
        int [] arr = new int [n];

        for (int i = 0; i < n - 1; i++) {
            int minIndex = i;

            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[minIndex]) {
                    minIndex = j;
                }
            }

            int temp = arr[i];
            arr[i] = arr[minIndex];
            arr[minIndex] = temp;
        }

        for (int num : arr) {
            System.out.print (num + " ");
        }
    }
}
```

LinearSearch.java — SearchSort — Visual Studio Code

File Edit Selection View Go Run Terminal Help

EXPLORER

▼ SEARCHSORT

LinearSearch.java

BinarySearch.java

BubbleSort.java

InsertionSort.java

SelectionSort.java

LinearSearch.java x

BinarySearch.java x

BubbleSort.java x

InsertionSort.java x

SelectionSort.java x

SearchSort > LinearSearch.java

```
1 // LinearSearch.java
2 // Searches for a target value by checking each element sequentially.
3
4 import java.util.Arrays;
5
6 public class LinearSearch{
7
8     /**
9      * Performs a linear search on the given array.
10     * @param arr the array to search
11     * @param target the value to find
12     * @return index of target, or -1 if not found
13     */
14     public static int linearSearch(int[] arr, int target) {
15         for(int i=0; i<arr.length; i++) {
16             if(arr[i]==target) {
17                 return i;
18             }
19         }
20         return -1;
21     }
22
23     public static void main(String[] args) {
24         int[] data={5,3,8,1,9,2,7,4,6};
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

bash

...

A

user@LAPTOP:~/SearchSort\$ javac LinearSearch.java user@LAPTOP:~/SearchSort\$ java LinearSearch Array: [5, 3, 8, 1, 9, 2, 7, 4, 6] Searching for: 7 Found 7 at index: 6 Elements checked: 7 user@LAPTOP:~/SearchSort\$

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Next Program

main 0 0 0 Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Java Prettier

BinarySearch.java — SearchSort — Visual Studio Code

File Edit Selection View Go Run Terminal Help

EXPLORER

- SEARCHSORT
 - LinearSearch.java
 - BinarySearch.java
 - BubbleSort.java
 - InsertionSort.java
 - SelectionSort.java

SearchSort · BinarySearch.java

```
1 // BinarySearch.java
2 // Searches a sorted array by repeatedly halving the search interval.
3
4 import java.util.Arrays;
5
6 public class BinarySearch{
7
8     /**
9      * Performs an iterative binary search.
10     * @param arr sorted array to search
11     * @param target value to locate
12     * @return index of target, or -1 if not found
13     */
14     public static int binarySearch(int[] arr, int target) {
15         int low = 0;
16         int high = arr.length - 1;
17
18         while (low <= high) {
19             int mid = low + (high - low) / 2;
20
21             if (arr[mid] == target) {
22                 return mid;
23             } else if (arr[mid] < target) {
24                 low = mid + 1;
25             }
26         }
27         return -1;
28     }
29 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

user@LAPTOP:~/SearchSort\$ javac BinarySearch.java user@LAPTOP:~/SearchSort\$ java BinarySearch Sorted Array: [1, 3, 5, 7, 9, 11, 13, 15, 17] Target: 11 Found 11 at index: 5 user@LAPTOP:~/SearchSort\$

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main 0 0 0 Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Java Prettier

BubbleSort.java — SearchSort — Visual Studio Code

File Edit Selection View Go Run Terminal Help

EXPLORER

▼ SEARCHSORT

- LinearSearch.java
- BinarySearch.java
- BubbleSort.java**
- InsertionSort.java
- SelectionSort.java

SearchSort > BubbleSort.java

```
1 // BubbleSort.java
2 // Repeatedly swaps adjacent elements that are in the wrong order.
3
4 import java.util.Arrays;
5
6 public class BubbleSort{
7
8     /**
9      * Sorts the array in ascending order using bubble sort.
10     * @param arr the array to sort (modified in-place)
11     */
12     public static void bubbleSort(int[] arr) {
13         int n = arr.length;
14         boolean swapped;
15
16         for(int i = 0; i < n - 1; i++) {
17             swapped = false;
18
19             for(int j = 0; j < n - i - 1; j++) {
20                 if(arr[j] > arr[j+1]) {
21                     int temp = arr[j];
22                     arr[j] = arr[j+1];
23                     arr[j+1] = temp;
24                     swapped = true;
25                 }
26             }
27         }
28     }
29 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

bash

```
user@LAPTOP:~/SearchSort$ javac BubbleSort.java user@LAPTOP:~/SearchSort$ java BubbleSort Before: [64, 34, 25, 12, 22, 11, 90] After: [11, 12, 22, 25, 34, 64, 90] user@LAPTOP:~/SearchSort$
```

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main 0 0 0 Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Java Prettier

InsertionSort.java — SearchSort — Visual Studio Code

File Edit Selection View Go Run Terminal Help

EXPLORER

▼ SEARCHSORT

- LinearSearch.java
- BinarySearch.java
- BubbleSort.java
- InsertionSort.java**
- SelectionSort.java

SearchSort > InsertionSort.java

```
1 // InsertionSort.java
2 // Builds a sorted array one element at a time by inserting each
3 // element into its correct position among already-sorted elements.
4
5 import java.util.Arrays;
6
7 public class InsertionSort{
8
9     /**
10      * Sorts an array in ascending order using insertion sort.
11      * @param arr the array to sort (modified in-place)
12      */
13     public static void insertionSort(int[] arr) {
14         int n = arr.length;
15
16         for(int i = 1; i < n; i++) {
17             int key = arr[i];
18             int j = i - 1;
19
20             // Shift elements greater than key one position ahead
21             while(j >= 0 && arr[j] > key) {
22                 arr[j+1] = arr[j];
23                 j--;
24             }
25         }
26     }
27 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

user@LAPTOP:~/SearchSort\$ javac InsertionSort.java user@LAPTOP:~/SearchSort\$ java InsertionSort Before: [12, 11, 13, 5, 6, 7] After: [5, 6, 7, 11, 12, 13] user@LAPTOP:~/SearchSort\$

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main 0 0 0 Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Java Prettier

SelectionSort.java — SearchSort — Visual Studio Code

File Edit Selection View Go Run Terminal Help

EXPLORER

- SEARCHSORT
 - LinearSearch.java
 - BinarySearch.java
 - BubbleSort.java
 - InsertionSort.java
 - SelectionSort.java

SearchSort › SelectionSort.java

```
1 // SelectionSort.java
2 // Divides the array into sorted and unsorted regions, then repeatedly
3 // selects the minimum element from the unsorted region.
4
5 import java.util.Arrays;
6
7 public class SelectionSort{
8
9     /**
10      * Sorts an array in ascending order using selection sort.
11      * @param arr the array to sort (modified in-place)
12      */
13     public static void selectionSort(int[] arr) {
14         int n = arr.length;
15
16         for (int i = 0; i < n - 1; i++) {
17             // Find the minimum element in the unsorted region
18             int minIdx = i;
19
20             for (int j = i + 1; j < n; j++) {
21                 if (arr[j] < arr[minIdx]) {
22                     minIdx = j;
23                 }
24             }
25         }
26     }
27 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

user@LAPTOP:~/SearchSort\$ javac SelectionSort.java user@LAPTOP:~/SearchSort\$ java SelectionSort Before: [29, 10, 14, 37, 13] After: [10, 13, 14, 29, 37] user@LAPTOP:~/SearchSort\$

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main 0 0 Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Java Prettier